

Multimodal Sentiment Analysis Based on Heterogeneous Contrastive Representation Learning with Prior Knowledge Enhancement

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Abstract—Multimodal sentiment analysis leverages various information sources (such as text, audio, video, etc.) to obtain more comprehensive and accurate sentiment information. Recent research has mainly focused on developing fusion mechanisms to bridge the modality gap, but there has been insufficient exploration into the heterogeneity of multimodal data, particularly in modeling the consistency and complementarity features between modalities, as well as the lack of guidance from prior knowledge in the fusion of heterogeneous data. In this paper, we propose a Heterogeneous Contrastive Representation Learning (HCRL) approach for multimodal sentiment analysis, which incorporates both consistency contrastive representation learning and complementarity semantic interactions. This approach allows for the effective learning of inter-modality similarity and difference information. Additionally, by incorporating prior knowledge to enhance the semantic interactions between multimodal features, the model enhances its capability to represent multimodal sentiment information. Experimental results on publicly available datasets CMU-MOSI and CMU-MOSEI demonstrate that our model outperforms state-of-the-art methods, showcasing the effectiveness of our proposed method.

Keywords—Multimodal sentiment analysis, Multimodal fusion, Representation learning, Prior knowledge

I. INTRODUCTION

Sentiment analysis plays a crucial role in natural language processing. Early sentiment analysis was mainly based on text data [1], aiming at analyzing, processing, summarizing, and inferring emotion in text data. In the context of the rapid increase in multimodal data over recent years [2], primarily driven by the swift advancements in both social media and the short video industry, there has been a discernible shift towards analyzing the surge of multimodal data and how people express their opinions and emotions using these diverse data types. Increasingly, people are opting to use visual and oral mediums, such as videos or images, to articulate and receive emotional information. Multimodal sentiment analysis (MSA) achieves enhanced precision in sentiment prediction by leveraging extensive data from multiple modalities [3], adeptly addressing the diverse modes of expression prevalent in the contemporary digital landscape. The complementary nature of multimodal data leads to more robust and comprehensive analysis results.

MSA faces the critical challenge of effectively fusing emotional information from different modalities to achieve

joint feature representation. Due to the deep research foundation and mature methods in the text modality, some scholars believe that the text modality is a key component in the multimodal fusion of MSA. Sun et al. [4] designed the Interaction Canonical Correlation Network (ICCN) based on multimodal data. Wang et al. [5] proposed a novel approach that learns paired cross-modal mappings for text, aiming to obtain effective unified multimodal representations. These studies emphasize the importance of the textual modality, yet fully utilizing its dominance in multimodal fusion requires further exploration. Another approach advocates studying the interaction and fusion between modalities by transforming from one modality to another. MCTN [6] utilizes transformations between different modalities to facilitate information fusion. MulT [7] employs attention blocks in cross-modal transformers, directly focusing on low-level features of other modalities without the need for alignment assumptions. CJTF [8] designed a coordinated joint translation fusion module to capture both common and private features. MTSA [9] uses modality translation for multimodal sentiment analysis, but MTSA overlooks heterogeneous information between modalities. TETFN [5] preserves heterogeneous information through unimodal predictions, and this approach may not fully capture all aspects of heterogeneity, potentially leading to a decline in model performance. Extracting consistent and complementary information between modalities during modal transformation has become a complex problem that must be addressed. Moreover, effectively incorporating prior knowledge to enhance the fusion process is crucial, as it can significantly enhance the quality of the learned representations by providing additional context and improving the alignment between different modalities.

Based on the above motivation, we propose a novel multimodal sentiment analysis framework called Heterogeneous Contrastive Representation Learning (HCRL). We posit that multimodal interactions exhibit two distinctive characteristics. The first is consistency features, which involve jointly describing entities to enhance textual semantics. The second is complementary features, encompassing unique information that harmonizes to supplement other modalities. Therefore, we design a text-centric multi-modal contrastive representation learning module based on modal conversion: the consistency contrastive representation learning module converts visual and auditory common semantic features into text features to enhance text features. The complementarity

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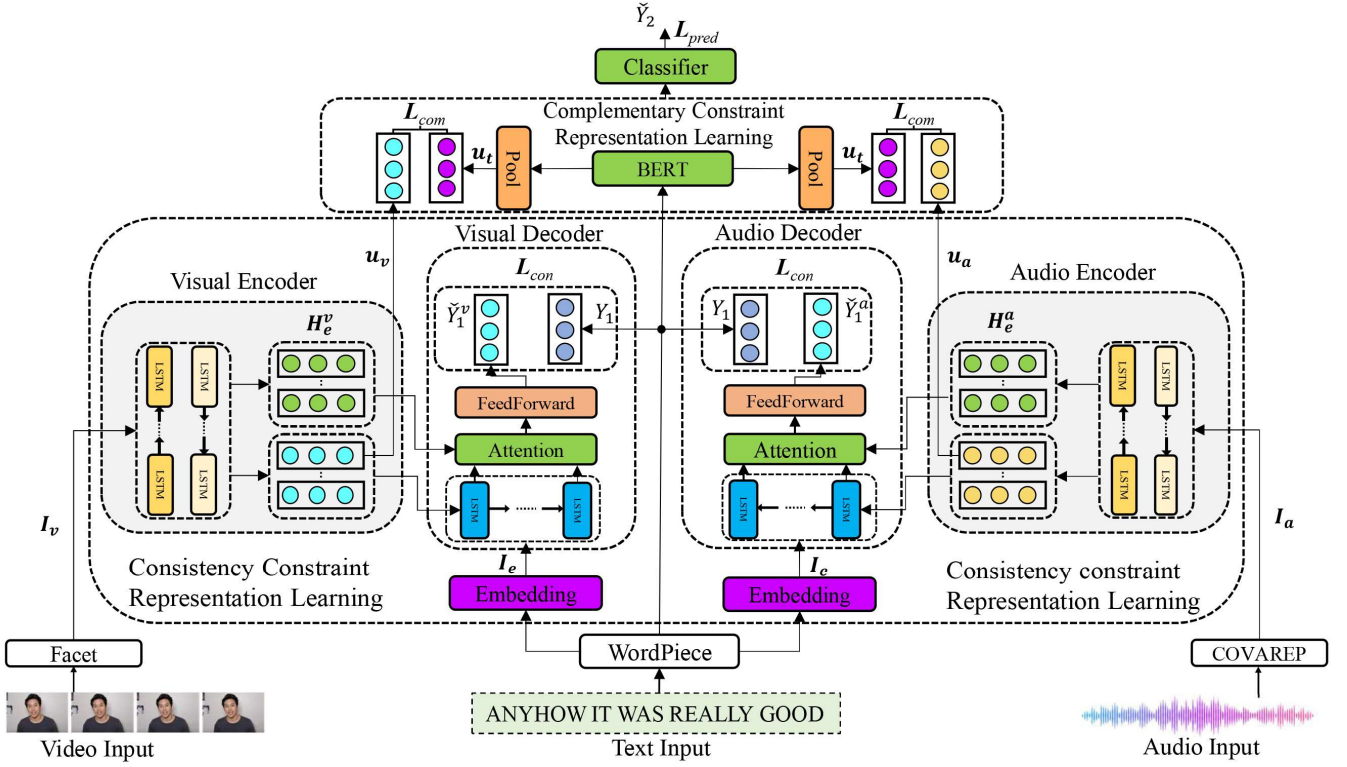


Fig. 1. The overall architecture of our proposed Heterogeneous Contrastive Representation Learning (HCRL).

contrastive representation learning module complements the text features by contrasting visual and auditory heterogeneous semantic features with text features. These two modules enhance the robustness of the model while taking advantage of the text modality. In addition, to extract emotional cues during the multimodal fusion process, we design a prior knowledge enhancement module. Prior knowledge is shown to be helpful in the process of single-modal sentiment analysis [10]. Therefore, we leverage prior knowledge, such as part-of-speech tags and polarity tags, to enhance the fusion process and enhance the model's perception of emotional knowledge. By incorporating these techniques, our HCRL framework comprehensively and accurately captures emotional information from multiple modalities, leading to improved sentiment analysis performance.

We evaluate the effectiveness of our proposed method on two widely recognized benchmark datasets, CMU-MOSI [11] and CMU-MOSEI [12]. The experimental results demonstrate that our proposed framework achieves state-of-the-art (SOTA) performance on both datasets.

To summarize, the main contributions of this paper are as follows:

- We propose a new multimodal sentiment analysis framework HCRL, using consistency and complementarity contrastive representation learning to build a cross-modal semantic interaction model to capture consistency features and complementarity features between modalities.
- A priori knowledge enhancement module is designed, using part-of-speech tags and word-level polarity tags as the expansion of sentiment knowledge, which strengthens the model's perception of emotional knowledge.

- We conduct experiments on two multimodal benchmark datasets, CMU-MOSI and CMU-MOSEI. Experimental results show that our proposed model achieves state-of-the-art performance on both datasets.

II. RELATED WORK

A. Multimodal Sentiment Analysis

In recent years, the rapid growth of the new media industry has led to a surge in the number of short videos, people are no longer limited to using just text to express emotions. Moreover, multi-modal sentiment analysis goes beyond single-modal approaches by integrating information from different modalities, as a result, sentiment analysis methods driven by multimodal data have garnered significant attention from researchers [13-14]. In terms of fusion strategy, multimodal sentiment analysis can be categorized into early fusion and late fusion. Early fusion involves simple concatenation of all modal features, which are then used as input for prediction models such as Hidden Markov Models (HMM) or Support Vector Machines (SVM). With the advent of deep learning, approaches like MKL [15] utilize deep convolutional networks, recurrent neural networks (RNN), and openAIR to extract visual, textual, and acoustic features, which are concatenated and fed into prediction models for classification. In late fusion, each modality is processed by separate prediction models, and the resulting predictions are averaged, weighted [16], or majority voted [17] to obtain the final prediction.

With the transformer structure proposed by Google [18], pre-trained language model develops rapidly. The pre-training model BERT [19] uses a new Masked Language Model (MLM) to generate deep bidirectional language representations. Increasingly, researchers have chosen to use pre-trained models directly as feature extractors for various

modalities in sentiment analysis tasks. For example, MISA [20] and Self-MM [21] replace traditional word embeddings like GloVe [22] with BERT to extract contextualized features from the original sentence. As the demand for multimodal data analysis increases, some multimodal pre-training models emerge in an endless stream, such as ViLBERT [23] and VL-BERT [24], they are pre-trained on large-scale datasets to solve various downstream tasks. Leveraging pre-trained models for multimodal sentiment analysis has now become a major trend.

In our approach, we use a pre-trained BERT model to extract discourse-level representations from the text modality. Unlike previous works, we convert visual and auditory features into text-centered representations, and based on this, we construct contrastive representation learning to extract consistency and complementarity features.

B. Prior Knowledge

In the realm of sentiment analysis, prior knowledge plays a crucial role in enhancing the effectiveness of sentiment analysis. Common forms of emotional knowledge include part of speech, emotional polarity, and knowledge graphs. These linguistic insights are often utilized as external features in sentiment analysis, contributing to the overall analysis process. By leveraging such prior emotional knowledge, sentiment analysis models can achieve improved performance and better capture the nuances of emotional content in texts. Based on the recent pre-trained BERT language model, Tian et al. [25] introduced emotional elements such as evaluation objects and emotional words in the text into the Mask Language Model pre-training task. Further improves the performance of BERT-like models on multiple sentiment classification datasets. Socher et al. [26] proved that part-of-speech helps in analyzing the syntactic structure of the text. Huang et al. [27] argued that parts-of-speech can help learn favorable representations of emotions.

C. Multimodal Feature Extraction

Visual feature extraction: After the emergence of deep learning, some classic visual feature extraction models have been proposed. In 2014, 2D-CNN proposed by You et al. In 2015, Tran et al. proposed 3D-CNN. In 2016, the convolutional recurrent neural network proposed by Poria et al. combines CNN and RNN. In the past few years, many sentiment analysis models either use 3D-CNN to extract visual features from videos, or use publicly available libraries to extract visual features, such as FACET, OKAO, and CERT libraries. With the emergence of the pre-training model ViT [28], it has also become the choice of many people. In our paper, video feature extraction uses the publicly available FACET library.

Audio Feature Extraction: Most recent multimodal sentiment analysis models use Open SMILE, COVAREP, Open EAR to extract acoustic features. In this paper, the COVAREP [29] library is used to extract audio features.

Text feature extraction: Most of the previous multimodal sentiment analysis used Glove or word2vec for word embedding. With the emergence of large pre-trained language models such as BERT, many articles directly use BERT as a text feature extractor. The text feature representation in this paper also uses BERT as the extractor.

III. METHODOLOGY

In this section, we elaborate on the details of the proposed Heterogeneous Contrastive Representation Learning (HCRL). The overall architecture of the model is shown in Fig. 1. The text encoder module, consistency and complementarity contrastive representation learning module and prior knowledge enhancement module are mainly introduced. In the contrastive representation learning module, the processing of video and audio is the same, we only introduce the processing of video modality.

A. Text Encoder

For text modal input, we first use NLTK, a natural language processing toolkit, to obtain its part-of-speech tagging (POS) and word-level polarity tagging, respectively. The joint input I_t composed of prior knowledge and text input, and then I_t is first tokenized by WordPiece for generating identifiable tokens. After that, these token ids are fed into the pre-trained BERT model to obtain the hidden state of the last layer of the first token (classification token) of the sequence, and this vector is used as the utterance-level representation u_t . In addition, these tokens ids go through Embedding to get the word embeddings I_e . I_e will be used as the input of the decoder to participate in the decoding of video and audio modality.

$$u_t = \text{Linear}(\text{BERT}(I_t)) \quad (1)$$

$$I_e = \text{Embedding}(I_t) \quad (2)$$

Where $u_t \in R^h$ and $I_e \in R^{l \times b}$, h is the output length of the linear layer, in order to make the speech-level representation projections the same length. b is the dimension of word embeddings in BERT. The linear layer in Equation 1 is to map the utterance-level representations of the three modalities into the same dimension. In addition, u_t will be passed into the complementarity contrastive representation learning module to participate in the complementarity representation learning.

B. Consistency and Complementarity Contrastive Representation Learning Module

On the encoder side of the consistency contrastive representation learning module is a BiLSTM network. The video feature I_v extracted by Facet is finally input into the network, which can be observed in Fig. 1. The initial hidden state of the encoder is $\overrightarrow{H_e}$ and $\overleftarrow{H_e}$, where $\overrightarrow{H_e} \in R^h$, $\overleftarrow{H_e} \in R^h$, h is the size of the encoder hidden layer. The input in each direction is denoted as $\overrightarrow{I_v}$, $\overleftarrow{I_v}$, the computation for $s=1, 2, \dots, l$ step is as follow:

$$\begin{aligned} \overrightarrow{H_e^s}, \overrightarrow{C_e^s} &= \text{LSTM}(\overrightarrow{I_v^s}, \overrightarrow{H_e^{s-1}}, \overrightarrow{C_e^{s-1}}), \\ \overleftarrow{H_e^{l-s+1}}, \overleftarrow{C_e^{l-s+1}} &= \text{LSTM}(\overleftarrow{I_v^{l-s+1}}, \overleftarrow{H_e^{l-s}}, \overleftarrow{C_e^{l-s}}), \\ H_e^s &= \frac{1}{2}(\overrightarrow{H_e^s} + \overleftarrow{H_e^{l-s+1}}) \end{aligned} \quad (3)$$

where $\overrightarrow{I_v^s}$ and $\overleftarrow{I_v^{l-s+1}}$ is the s -th entry of I_v . To make the most of contextual information, we use the last output of each direction to compute the utterance-level representation, that is $u_v = \frac{1}{2}(\overrightarrow{H_e^l} + \overleftarrow{H_e^1})$, where $u_v \in R^h$. The generated

multimodal utterance-level representations are guaranteed to have the same length.

$$\tilde{Y}_1, H_d = \text{dec}(I_e, H_e) \quad (4)$$

Where I_e is the text embedding, H_e is the output of encoder, $\tilde{Y}_1$ is the output of decoder, it represents the result of modal translation. We use $\tilde{Y}_1^s$ to denote the prediction of the (s+1)-th word in I_t , and it will be used to pass to the loss function to participate in the consistency contrastive representation learning. Specifically, given the hidden states of the encoder and the decoder across all the time steps by $H_e \in R^{l \times h}$ and $H_d \in R^{l \times h}$, the Query, Key and Value are defined as $Q = H_d$, $K = V = H_e$.

Our objective function consists of three different tasks: a consistency contrastive representation learning task, a complementarity contrastive representation learning task, and a sentiment prediction task. For the consistency contrastive representation learning task, the prediction is given by $\tilde{Y}_1^s$ and the truth label is the one-hot version of the token id. The truth label is recorded as Y_1^s . We use cross-entropy to measure the loss between the above prediction and the true label. The loss is computed as follows:

$$L_{con} = - \sum_{s=1}^{l-1} (Y_1^s \log(\tilde{Y}_1^s) + (1 - Y_1^s) \log(1 - \tilde{Y}_1^s)) \quad (5)$$

The complementarity contrastive representation learning uses the hidden state u_v of the last layer of LSTM in the encoder and the previously acquired u_t to capture difference information. For the complementarity contrastive representation learning task, we design the loss function to widen the distance between u_v and u_t , mainly using the cosine similarity between them. The lower the similarity between two vectors, the smaller the loss, to capture complementary emotional information. Calculated as follows:

$$L_{com} = \frac{\sum_{i=1}^h u_{v_i} * u_{t_i}}{\sqrt{\sum_{i=1}^h (u_{v_i})^2} * \sqrt{\sum_{i=1}^h (u_{t_i})^2}} \quad (6)$$

For the sentiment prediction task, we use Mean Squared Error (MSE) to measure the loss. Given the prediction and true label by $\tilde{Y}_2$ and Y_2 , where $Y_2 \in [-3, 3]$. The computation is as follows:

$$L_{pred} = \frac{1}{n} \sum_{i=1}^n (\tilde{Y}_{2i} - Y_{2i})^2 \quad (7)$$

Both the consistency contrastive loss and the complementarity contrastive loss are composed of two parts, which come from the video modality and the audio modality respectively. In addition, a weighted summary is applied to integrate L_{con} , L_{com} , and L_{pred} . L_{total} represents our total loss, the computation is as follows:

$$L_{total} = w_1 L_{con} + w_2 L_{com} + w_3 L_{pred} \quad (8)$$

We set the parameter values of w_1 , w_2 , and w_3 according to the experimental experience to obtain the best effect. The values of w_1 , w_2 and w_3 are 1, 0.1 and 0.1, respectively.

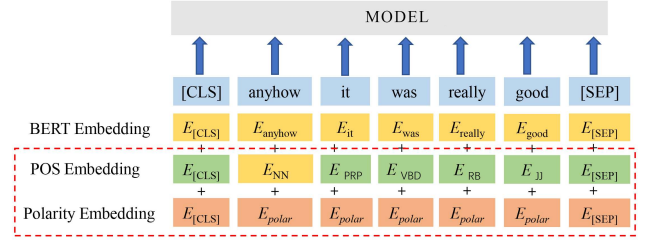


Fig. 2. Prior Knowledge Enhancement Module.

C. Prior Knowledge Enhancement Module

This module tags each word with part-of-speech tags and word-level polarity tags. The input of the module is a text sequence $X = (x_1, x_2, \dots, x_n)$, where x_i ($1 \leq i \leq n$) represents a word in the text sequence. First, we get the part-of-speech tag pos_i of each word x_i via `pos_tag`. Such as verbs(v), nouns (n), adjectives (adj), adverbs (adv), etc., and then use NLTK to obtain word-level emotional polarity, and the final expression is as follows $(x_i, pos_i, polar_i)$. The module structure is shown in Fig. 2. Incorporating part-of-speech tags and word-level polarity tags enriches textual features, helping the model understand grammatical and emotional contexts. This additional information improves sentiment identification and provides a structured, emotionally guided text representation for multimodal fusion. Consequently, it enhances feature alignment and interaction across modalities, boosting the precision and robustness of sentiment recognition.

IV. EXPERIMENTS

A. Datasets

CMU-MOSI dataset: This dataset includes 2199 emotion-annotated dialogues extracted from 93 videos collected from YouTube video blogs (vlogs). Each utterance in the dataset is labeled with one of the following sentiments: strongly positive (labeled as +3), positive (+2), weakly positive (+1), neutral (0), weakly negative (-1), negative (-2), strongly negative (-3).

CMU-MOSEI dataset: The dataset contains 23453 labeled video segments segmented from 2928 videos. The videos cover 250 different topics. Each utterance in the dataset is labeled with one of the following sentiments: strongly positive (labeled as +3), positive (+2), weakly positive (+1), neutral (0), weakly negative (-1), negative (-2), strongly negative (-3).

B. Evaluation Metrics

Similar to previous work, we use five evaluation metrics to measure the performance of the model on the CMU-MOSI and CMU-MOSEI datasets: They are Acc-7, Acc-2, F1 Score, Mean Absolute Error (MAE) and Pearson Correlation (Corr), respectively. Among them, Acc-2 and F1 are used to measure the accuracy of binary classification; Acc-7 measures the accuracy of the classification of seven categories from -3 to 3; Pearson correlation (Corr) is used to measure the correlation between estimated and predicted values; MAE is used to calculate the mean absolute error.

C. Feature Extraction

The experimental dataset in this paper involves three modalities, including text, audio and video. The feature extraction methods for the three modalities are as follows:

TABLE I. RESULTS OF SENTIMENT ANALYSIS ON CMU-MOSEI AND CMU-MOSI. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Dataset Metric	MOSI					MOSEI				
	Acc2	F1	MAE	Corr	Acc7	Acc2	F1	MAE	Corr	Acc7
RAVEN	78.0	76.6	0.915	0.691	33.2	79.1	79.5	0.614	0.662	50.0
MuT	83.0	82.8	0.871	0.698	40.0	82.5	82.3	0.580	0.703	51.8
Self-MM	86.0	86.0	0.713	0.798	-	85.2	85.3	0.530	0.765	-
MAG-BERT	86.1	86.0	0.712	0.796	-	84.7	84.5	0.624	0.769	-
MMIM	86.1	86.0	0.700	0.800	46.7	86.0	86.0	0.526	0.772	54.2
TETFN	86.1	86.1	0.717	0.800	-	85.2	85.3	0.551	0.748	-
CJTF	86.5	86.4	0.704	0.810	-	86.1	86.1	0.513	0.778	-
MTSA	86.8	86.8	0.696	0.806	46.4	85.5	85.3	0.541	0.774	52.9
HCRL(ours)	87.4	87.4	0.693	0.811	51.5	86.2	86.2	0.511	0.779	54.6

Video features: In this paper, the FACET library is used to extract visual features. We use FACET to extract a set of visual features, including facial action units, facial landmarks, head pose, gaze tracking and HOG features.

Audio features: In this paper, COVAREP [29] is used to extract acoustic features, including 12 Mel frequency cepstral coefficients, pitch and voiced/unvoiced segmentation features, glottal source parameters, peak slope parameters, and maximum dispersion quotient [30].

Text features: Youtube API transcription of the audio, and then directly use BERT to extract text features.

D. Training Settings

The model training uses AdamW optimizer, the initial learning rate is 1×10^{-5} , and the learning rate gradually decreases as the training progresses. For the encoder in the consistency contrastive representation learning module, we use a bidirectional LSTM with a hidden size set to 256 and a dropout of 0.3. Decoder parameters are similar. We use the pre-trained bert-base-uncased model in the Transformers library, the hidden state of the last layer of the first token (classification token) as the utterance-level representation of the text, which means hidden_size=768. Our training framework is pytorch1.6.0.

We run on a server equipped with 2 Tesla V100 GPUs, the CMU-MOSI dataset achieves the best result on the 35th epoch, and the CMU-MOSEI dataset achieves the best result on the 7th epoch.

E. Baselines

In order to fully guarantee the effectiveness of HCRL, we compare the performance of our model with the following baseline and state-of-the-art models.

RAVEN: Recursive Attentional Variation Embedding Network (RAVEN) [31] learns fine-grained structure of nonlinguistic subword sequences and dynamically changes word representations based on nonlinguistic cues.

MuT: Multimodal Transformer (MuT) [7] uses attention blocks in cross-modal transformers to directly focus on low-level features of other modalities without alignment assumptions.

Self-MM: Learning Modality-Specific Representations with Self-Supervised Multi-Task Learning (Self-MM) [21] designs a multi- and a uni-task to learn inter-modal consistency and intra-modal specificity.

TABLE II. ABLATION STUDIES ON MOSI.

	Acc2	F1	MAE	Corr	Acc7
HCRL	87.4	87.4	0.693	0.811	51.5
HCRL w/o PM	87.2	87.3	0.711	0.794	50.7
HCRL w/o CM	86.9	86.9	0.710	0.784	49.3
HCRL w/o CM&PM	86.8	86.8	0.698	0.806	46.4

MAG-BERT: Bert's Multimodal Adaptation Gate (MAG-Bert) [32] combines aligned non-linguistic information with text representations in Bert.

MMIM: MultiModal InfoMax (MMIM) [33] hierarchically maximizes the Mutual Information (MI) in unimodal input pairs (inter-modality) and between multimodal fusion result and unimodal input in order to maintain task-related information through multimodal fusion.

TETFN: A novel approach called Text Enhanced Transformer Fusion Network (TETFN) [5] is proposed that learns text-oriented pairwise cross-modal mappings for efficient unified multimodal representations.

CJTF: Coordinated Joint Translation Fusion (CJTF) [8] integrates public and private features of non-textual information to enhance and complement textual features containing semantics and syntactics from sentiment-interactive GCN.

MTSA: Multimodal Translation for Sentiment Analysis (MTSA) [9], a multimodal framework that improves visual and the quality of the audio characteristics.

F. Experimental Results

The results of the sentiment analysis on CMU-MOSI and CMU-MOSEI are shown in Table I. In general, our HCRL framework achieves the best performance on both datasets. Compared with the suboptimal method, on the MOSI dataset, HCRL achieved improvements of 0.6% on Acc2 and F1, and 4.8% in Acc7. And on the MOSEI dataset, all indicators improved slightly. This demonstrates the effectiveness of our method.

V. ABLATION STUDIES

Our model has two important modules: the complementarity contrastive representation learning module and the prior knowledge guidance module. To verify the roles of these two modules, we conducted several ablation experiments on the CMU-MOSI dataset. The results of the

study are shown in Table II. "w/o" indicates the removal of a specific module from the proposed model. CM and PM represent the complementarity contrastive representation learning module and the prior knowledge guidance module, respectively. The performance degrades to varying degrees when these components are removed. Additionally, removing the consistency contrastive representation learning module would reduce the model to single-modal sentiment analysis, which is inconsistent with our multimodal approach, so we did not conduct ablation experiments on the consistency module.

VI. CONCLUSION

This paper aims to explore new multimodal fusion methods and overcome the limitations of existing multimodal sentiment analysis approaches. We propose a new framework, HCRL, which adopts consistency and complementarity contrastive representation learning to model cross-modal semantic interactions. The framework extracts information about the consistency and complementarity between textual and non-textual modalities, enhancing the robustness of the model. Additionally, we use prior knowledge to improve the model's perception of emotional knowledge and strengthen the text modality. Experiments on the CMU-MOSI and CMU-MOSEI datasets demonstrate that HCRL outperforms existing state-of-the-art methods across multiple metrics. In the future, we will continue to explore the fusion mechanisms between modalities and develop more effective fusion methods.

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